



SAPIENZA
UNIVERSITÀ DI ROMA

Advanced Spacecraft Dynamics

Homework 4

2025

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12/05/2025

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1 Study-case Statement

The purpose of the study is to investigate minimum-time continuous-thrust transfer from two orbits. In particular:

- Initial orbit:
 - Equatorial
 - Semi-major axis $a_0 = 10000$ km
 - Eccentricity $e_0 = 0.3$
- Final orbit:
 - Equatorial
 - Circular
 - Radius $r_f = 42164$ km

The thrust system is characterized by the exhaust velocity $c = 30$ km and maximum acceleration, with respect to the initial mass, $u_T^{max} = 0.01 g_0$, where $g_0 = 9.8066 m/s^2$ is the reference gravity.

The indirect optimization and Pontryagin's Minimum Principle will be used.

2 Mathematical Model

This chapter presents the mathematical framework developed to address the optimization problem outlined in Chapter 1.

First, nondimensionalization is performed in order to rescale all physical variables into dimensionless, order-of-magnitude unity quantities. By normalizing variables, the condition number of the governing equations is improved, thereby enhancing the stability and convergence of numerical solvers. This step is critical for iterative methods like Newton-type algorithms or gradient-based optimization routines, which often struggle with ill-conditioned systems.

Second, we employ an indirect optimization method to reformulate the constrained optimization problem as a Two-Point Boundary-Value Problem (TPBVP). Unlike direct

methods that discretize the problem upfront, indirect approaches leverage calculus of variations to derive necessary optimality conditions. These conditions manifest as a system of differential equations with boundary constraints at both initial and final states.

Finally, Pontryagin's Minimum Principle (PMP) is applied, to identify the optimal control strategy. By minimizing the Hamiltonian function the optimal control law is derived, yielding a solution that satisfies both the dynamics and the problem constraints.

2.1 Nondimensionalization

The reference scales for nondimensionalization are defined as follows:

$$\begin{array}{ll} DU = 6378.136 \text{ km} & \text{Distance Unit} \\ TU = 806.81 \text{ s;} & \text{Time Unit} \\ m_0 & \text{Initial Mass} \end{array}$$

They are chosen so that the gravitational parameter of the Earth is unitary in DU and TU . In fact:

$$\mu = 398600 \text{ km}^3/\text{s}^2 = \frac{398600 \text{ km}^3/\text{s}^2}{DU^3/TU^2} \frac{DU^3}{TU^2} = 1 \frac{DU^3}{TU^2}$$

The parameters defined in the previous chapter read:

$$\begin{aligned} a_0 &= 10000 \text{ km} = \frac{10000 \text{ km}}{DU} DU = 1.5679 DU \\ e_0 &= 0.3 \\ r_f &= 42164 \text{ km} = \frac{42164 \text{ km}}{DU} DU = 6.6107 DU \\ c &= 30000 \text{ km/s} = \frac{30000 \text{ km/s}}{DU/TU} \frac{DU}{TU} = 3.7949 \frac{DU}{TU} \\ g_0 &= 9.8066 \text{ m/s}^2 = \frac{9.8066 \times 10^{-3} \text{ km/s}^2}{DU/TU^2} \frac{DU}{TU^2} = 1.0008 \frac{DU}{TU^2} \end{aligned}$$

2.2 Indirect Optimization

The dynamical system is described by $n = 5$ variables, which form the state vector $\underline{x}$. In this case:

$$\underline{x} = [r \quad \theta^* \quad v_r \quad v_t \quad \xi] \quad (1)$$

where: r is the radius (distance from the center of the Earth to the spacecraft); θ^* is the true anomaly; v_r and v_t are respectively the radial and transverse component of the velocity vector; $\xi := m/m_0$ is the mass ratio.

The dynamical system is controlled by $m = 2$ time-varying variables, which form the control vector:

$$\underline{u} = [u_T \quad \phi] \quad (2)$$

where: $u_T = T/m_0$ is the acceleration provided by the trust with respect to the initial mass; ϕ is the direction of the thrust in the orbital plane, defined as the angle between the thrust and the radial direction (see Figure 1).

One time-independent parameter is also considered:

$$p = \theta_0^* \quad (3)$$

where θ_0^* is the initial true anomaly, to be optimized. For the sake of generality, the parameter will be referred to as a vector of parameters $\underline{p}$ since, in other cases, multiple time-independent variables may be present.

For clarity, the reference frames used and the related rotation matrices are explicitly defined. The Local-Horizontal Local-Vertical (LHLV) frame is depicted in Figure 1 and the body-fixed commanded frame is depicted in Figure 2. The following relations among the inertial reference frame, the LHLV reference frame and the body-fixed commanded frame holds:

$$\begin{bmatrix} \hat{r} \\ \hat{\theta} \\ \hat{h} \end{bmatrix} = R_3(\omega + \theta^*) \begin{bmatrix} \hat{c}_1 \\ \hat{c}_2 \\ \hat{c}_3 \end{bmatrix} \quad (4)$$

$$\begin{bmatrix} \hat{b}_{c1} \\ \hat{b}_{c2} \\ \hat{b}_{c3} \end{bmatrix} = R_1(270) R_3(\phi) \begin{bmatrix} \hat{r} \\ \hat{\theta} \\ \hat{h} \end{bmatrix} \quad (5)$$

$$\Rightarrow \begin{bmatrix} \hat{b}_{c1} \\ \hat{b}_{c2} \\ \hat{b}_{c3} \end{bmatrix} = R_1(270) R_3(\omega + \theta^* + \phi) \begin{bmatrix} \hat{c}_1 \\ \hat{c}_2 \\ \hat{c}_3 \end{bmatrix} \quad (6)$$

where: $R_i(\alpha)$ is the elementary rotation of angle α about axis i ; ω is the argument of perigee, null for this study-case.

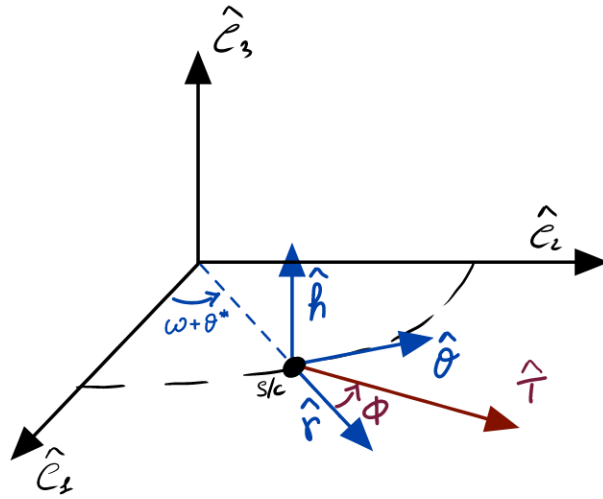


Figure 1: The Local-Horizontal Local-Vertical (LHLV) frame within the inertial reference frame.

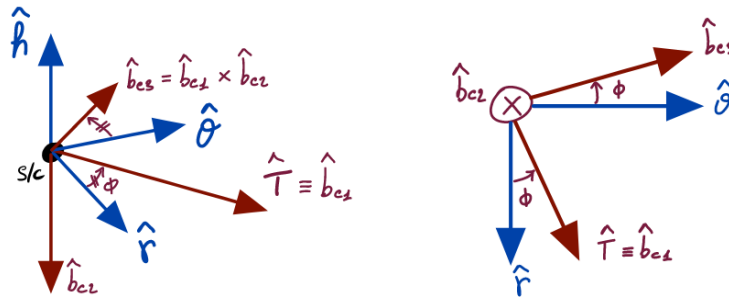


Figure 2: the body-fixed commanded frame within the LHLV frame.

The dynamical equations $\dot{\underline{x}} = \underline{f}(\underline{x}, \underline{u}, t)$ are written in the LHLV reference frame as follows:

$$\begin{cases} \dot{r} &= v_r \\ \dot{\theta}^* &= \frac{v_t}{r} \\ \dot{v}_r &= \frac{v_t^2}{r} - \frac{\mu}{r^2} + \frac{u_T}{\xi} \cos \phi \\ \dot{v}_\theta &= -\frac{v_r v_t}{r} + \frac{u_T}{\xi} \sin \phi \\ \dot{\xi} &= -\frac{u_T}{c} \end{cases} \quad (7)$$

The boundary conditions $\underline{\psi}(\underline{x}_0, \underline{x}_f, \underline{p}, t_0, t_f) = 0$ read:

$$0 = r(0) - \frac{p_0}{1 + e_0 \cos \theta_0^*} \quad (8)$$

$$0 = \theta^*(0) - \theta_0^* \quad (9)$$

$$0 = v_r(0) - \sqrt{\frac{\mu}{p_0}} e_0 \sin \theta_0^* \quad (10)$$

$$0 = v_t(0) - \sqrt{\frac{\mu}{p_0}} (1 + e_0 \cos \theta_0^*) \quad (11)$$

$$0 = \xi(0) - 1 \quad (12)$$

$$0 = r(t_f) - r_f \quad (13)$$

$$0 = v_r(t_f) - 0 \quad (14)$$

$$0 = v_t(t_f) - \sqrt{\frac{\mu}{r_f}} \quad (15)$$

where: $p_0 = a_0(1 - e_0^2)$ is the semilatus rectum of the departure orbit. The initial true anomaly θ_0^* and the time of flight t_f represent the parameters to be optimized. All other initial conditions are then determined given a choice of θ_0^* and t_f .

The optimization process aims to minimize the following objective function J , expressed in the Mayer form:

$$J = t_f \quad (16)$$

A locally-minimizing optimal solution $\underline{x}^*(t)$, associated with $\underline{u}^*(t)$ and $\underline{p}^*$, is such that J is locally minimized, while holding the following constraints:

$$\underline{\dot{x}}^*(t) - \underline{f}(\underline{x}^*, \underline{u}^*, t) = 0 \quad \forall t \in [t_0, t_f] \quad (17)$$

$$\underline{\psi}(\underline{x}_0^*, \underline{x}_f^*, \underline{p}^*, t_0, t_f) = 0 \quad (18)$$

The Lagrange multiplier method is particularly well-suited for solving optimization problems that involve constraints. By introducing additional variables, known as Lagrange multipliers, this approach transforms a constrained problem into an unconstrained one, allowing the optimization to consider both the objective function and the constraint conditions simultaneously. The method ensures that the solution not only optimizes the desired performance index (minimum time of flight, for this study-case) but also adheres rigorously to the constraints. In particular, the time-varying adjoint vector $\underline{\lambda}(t)$ and the time-independent adjoint vector $\underline{\nu}$ are considered to build the extended objective function $\bar{J}$:

$$\bar{J} = t_f + \underline{\nu}^\top \underline{\psi} + \int_{t_0}^{t_f} \underline{\lambda}^\top (\underline{f} - \underline{\dot{x}}) dt \quad (19)$$

While constraints (17) and (18) holds true, the minimization of $\bar{J}$ is equivalent to the minimization of the original objective function J . By integrating by parts:

$$\bar{J} = t_f + \underline{\nu}^\top \underline{\psi} + \underline{\lambda}_0^\top \underline{x}_0 - \underline{\lambda}_f^\top \underline{x}_f + \int_{t_0}^{t_f} \left(\underline{\lambda}^\top \underline{f} + \dot{\underline{\lambda}}^\top \underline{x} \right) dt \quad (20)$$

It is convenient to define the Hamiltonian function:

$$H := \underline{\lambda}^\top \underline{f} = \lambda_r v_r + \lambda_{v_r} \left(\frac{v_t^2}{r} - \frac{\mu}{r^2} + \frac{u_T}{\xi} \cos \phi \right) + \lambda_{v_t} \left(-\frac{v_r v_t}{r} + \frac{u_T}{\xi} \sin \phi \right) - \lambda_\xi \frac{u_T}{c} \quad (21)$$

The extended objective functions reads:

$$\bar{J} = t_f + \underline{\nu}^\top \underline{\psi} + \underline{\lambda}_0^\top \underline{x}_0 - \underline{\lambda}_f^\top \underline{x}_f + \int_{t_0}^{t_f} \left(H + \dot{\underline{\lambda}}^\top \underline{x} \right) dt \quad (22)$$

A necessary condition for minimizing the extended objective function $\bar{J}$ is that:

$$d\bar{J} = 0 \quad (23)$$

By expanding each term, it reads:

$$\begin{aligned} d\bar{J} = & dt_f + \cancel{\underline{\psi}} + \underline{\nu}^\top \left(\frac{\partial \psi}{\partial \underline{p}} d\underline{p} + \frac{\partial \psi}{\partial \underline{x}_0} d\underline{x}_0 + \frac{\partial \psi}{\partial \underline{x}_f} d\underline{x}_f + \frac{\partial \psi}{\partial t_0} dt_0 + \frac{\partial \psi}{\partial t_f} dt_f \right) + \\ & + \underline{\lambda}_0^\top d\underline{x}_0 + \underline{x}_0^\top d\underline{\lambda}_0 - \underline{\lambda}_f^\top d\underline{x}_f - \underline{x}_f^\top d\underline{\lambda}_f + \left(H_f + \dot{\underline{\lambda}}_f^\top \underline{x}_f \right) dt_f - \left(H_0 + \dot{\underline{\lambda}}_0^\top \underline{x}_0 \right) dt_0 + \\ & + \int_{t_0}^{t_f} \left[\left(\frac{\partial H}{\partial \underline{x}} + \dot{\underline{\lambda}}^\top \right) \delta \underline{x} + \frac{\partial H}{\partial \underline{\lambda}} \delta \underline{\lambda} + \cancel{\frac{\partial H}{\partial \underline{p}} \delta \underline{p}} + \frac{\partial H}{\partial \underline{u}} \delta \underline{u} + \underline{x}^\top \delta \dot{\underline{\lambda}} \right] dt \end{aligned} \quad (24)$$

By integrating by parts:

$$\begin{aligned} d\bar{J} = & dt_f + \underline{\nu}^\top \left(\frac{\partial \psi}{\partial \underline{p}} d\underline{p} + \frac{\partial \psi}{\partial \underline{x}_0} d\underline{x}_0 + \frac{\partial \psi}{\partial \underline{x}_f} d\underline{x}_f + \frac{\partial \psi}{\partial t_0} dt_0 + \frac{\partial \psi}{\partial t_f} dt_f \right) + \\ & + \underline{\lambda}_0^\top d\underline{x}_0 + \underline{x}_0^\top d\underline{\lambda}_0 - \underline{\lambda}_f^\top d\underline{x}_f - \underline{x}_f^\top d\underline{\lambda}_f - \left(H_0 + \dot{\underline{\lambda}}_0^\top \underline{x}_0 \right) dt_0 + \\ & + \delta \dot{\underline{\lambda}}_f^\top \underline{x}_f - \delta \dot{\underline{\lambda}}_0^\top \underline{x}_0 + \int_{t_0}^{t_f} \left[\left(\frac{\partial H}{\partial \underline{x}} + \dot{\underline{\lambda}}^\top \right) \delta \underline{x} + \cancel{\frac{\partial H}{\partial \underline{\lambda}} \delta \underline{\lambda}} + \frac{\partial H}{\partial \underline{u}} \delta \underline{u} - \cancel{\delta \underline{\lambda}^\top \underline{x}} \right] dt \end{aligned} \quad (25)$$

where the definition (21) was used. By recalling that $d\underline{\lambda} = \delta \underline{\lambda} + \dot{\underline{\lambda}} dt$, the necessary condition for optimality reads, finally:

$$\begin{aligned} 0 = d\bar{J} = & dt_f + \underline{\nu}^\top \left(\frac{\partial \psi}{\partial \underline{p}} d\underline{p} + \frac{\partial \psi}{\partial \underline{x}_0} d\underline{x}_0 + \frac{\partial \psi}{\partial \underline{x}_f} d\underline{x}_f + \frac{\partial \psi}{\partial t_0} dt_0 + \frac{\partial \psi}{\partial t_f} dt_f \right) + \\ & + \underline{\lambda}_0^\top d\underline{x}_0 - \underline{\lambda}_f^\top d\underline{x}_f + H_f dt_f - H_0 dt_0 + \\ & + \int_{t_0}^{t_f} \left[\left(\frac{\partial H}{\partial \underline{x}} + \dot{\underline{\lambda}}^\top \right) \delta \underline{x} + \frac{\partial H}{\partial \underline{u}} \delta \underline{u} \right] dt \end{aligned} \quad (26)$$

The necessary condition must hold true for any values of differentials and variations. Therefore all the coefficient must vanish, leading to the complete set of first-order necessary conditions for optimality, casted to the study-case into account:

$$\dot{\underline{\lambda}}^\top = -\frac{\partial H}{\partial \underline{x}} \quad (27)$$

$$\frac{\partial H}{\partial \underline{u}} = \underline{0} \quad (28)$$

$$\underline{\nu}^\top \frac{\partial \psi}{\partial \underline{p}} = \underline{0} \quad (29)$$

$$\underline{\lambda}_0^\top = -\underline{\nu}^\top \frac{\partial \psi}{\partial \underline{x}_0} \quad (30)$$

$$\underline{\lambda}_f^\top = \underline{\nu}^\top \frac{\partial \psi}{\partial \underline{x}_f} \quad (31)$$

$$H_0 = \underline{\nu}^\top \frac{\partial \psi}{\partial t_0} \quad (32)$$

$$H_f = -\underline{\nu}^\top \frac{\partial \psi}{\partial t_f} - 1 \quad (33)$$

Equations (27), (28), known as Euler-Lagrange equations, together with equation (29) are a set of differential and algebraic equations whose solution satisfies the necessary conditions for optimality. They must hold true at all times. Equations (30) - (33) represents the boundary condition for the differential equations of the set. The system has to be integrated with the dynamical system (7) and the associated boundary conditions (8) - (15).

Some general rules can be outlined by the structures of equations (27) - (33):

- If the boundary condition of a state variable is not specified, then the associated adjoint variable is set to zero at that time, according to the adjoint boundary condition (e.g. if r_0 is not given, $\lambda_{r,0} = 0$). Conversely, if the boundary condition of a state variable is specified, then the associated adjoint variable is unknown at that time (e.g. if r_0 is given, $\lambda_{r,0} = -\nu_1$).
- If the initial (or final) time is not specified, the value of the Hamiltonian function at that time is set to zero. Conversely, if the time is specified, then the Hamiltonian function at that time is unknown.

Variation to the last rule occurs due to the minimization of the final time, that yields the condition $H_f = -1$. This condition can be relaxed by assuming $J^* = k_1 t_f$, where the constant $k_1 > 0$, such that minimizing J^* is equivalent to minimize J and the equation

(33) is turned into a inequality constrained to be assessed a posteriori:

$$H_f < 0 \quad (34)$$

2.3 Pontryagin Minimum Principle

Pontryagin Minimum Principle (PMP) represents a generalization of the condition (28), that accounts for possible constraint on the control $\underline{u} \in \mathcal{U}$. In particular, in this study-case the control is limited by $\underline{u} \in \mathcal{U} = [-u_T^{max}, u_T^{max}] \times (-\infty, \infty)$.

The control $\underline{u}$ must minimize the Hamiltonian function H :

$$\underline{u}(t) \in \arg \min_{\underline{u} \in \mathcal{U}} H(\underline{x}(t), \underline{\lambda}(t), \underline{u}). \quad (35)$$

The dependence of H on ϕ is here highlighted:

$$H \propto \lambda_{v_r} \frac{u_T}{\xi} \cos \phi + \lambda_{v_t} \frac{u_T}{\xi} \sin \phi \quad (36)$$

H is minimized if:

$$\lambda_{v_t} = -k_2 \sin \phi \quad \wedge \quad \lambda_{v_r} = -k_2 \cos \phi \quad (37)$$

where $k_2 = \sqrt{\lambda_{v_r}^2 + \lambda_{v_t}^2}$ is positive and constant. The above condition sets the optimal direction that the thrust must assume. In particular, the thrust $\underline{T}$ must be parallel to the prime vector, i.e. the vector defined as follows:

$$\underline{T} \parallel [-\lambda_{v_r} \quad -\lambda_{v_t}]^\top \quad (38)$$

The same condition can be reached by imposing the stationary condition (28) on H . In particular:

$$0 = \frac{\partial H}{\partial \phi} = -\lambda_{v_r} \frac{u_T}{\xi} \sin \phi + \lambda_{v_t} \frac{u_T}{\xi} \cos \phi \iff \begin{cases} \lambda_{v_t} = -k_2 \sin \phi \\ \lambda_{v_r} = -k_2 \cos \phi \end{cases} \quad (39)$$

By following a similar reasoning, it is possible to find the optimal value for the magnitude of the thrust. The dependence of H on u_T^{max} is here highlighted:

$$H \propto \lambda_{v_r} \frac{u_T}{\xi} \cos \phi + \lambda_{v_t} \frac{u_T}{\xi} \sin \phi - \lambda_\xi \frac{u_T}{c} = -\frac{u_T}{\xi} k_2 - \lambda_\xi \frac{u_T}{c} \quad (40)$$

where the condition (37) was used. Considering that $k_2 > 0$, if $\lambda_\xi > 0$, then H is minimized by imposing $u_T = u_T^{max}$. The condition $\lambda_\xi > 0$ must be proved. From the Euler-Lagrange equation (27), it follows:

$$\dot{\lambda}_\xi = -\frac{\partial H}{\partial \xi} = -\frac{u_T^{max}}{\xi^2} k_2 < 0 \quad (41)$$

In addition, since the final mass is not specified as boundary condition, according to equation (31), it follows that $\lambda_{f,0} = 0$. It can be concluded that $\lambda_\xi < 0 \quad \forall t \in [t_0, t_f]$.

Finally, the optimal control is given by:

$$\underline{u} = [u_T^{max} \quad \phi]^\top \quad \phi = \text{atan2}(-\lambda_{v_t}, -\lambda_{v_r}) \quad (42)$$

The Pontryagin Minimum Principle replaces equation (28).

3 Indirect Optimization: Two-Point Boundary-Value Problem

The indirect optimization problem has transformed the original problem of optimization with constraints into a Two-Point Boundary-Value Problem. The overall system to be integrated is here outlined:

$$\left\{ \begin{array}{lcl} \dot{r} & = & v_r, \\ \dot{\theta} & = & \frac{v_t}{r} \\ \dot{v}_r & = & \frac{v_t^2}{r} - \frac{\mu}{r^2} + \frac{u_T}{\xi} \cos \phi \\ \dot{v}_t & = & -\frac{v_r v_t}{r} + \frac{u_T}{\xi} \sin \phi \\ \dot{\xi} & = & -\frac{u_T}{c} \\ \dot{\lambda}_r & = & \lambda_{v_r} \frac{v_t^2 - 2\mu/r}{r^2} - \lambda_{v_t} \frac{v_r v_t}{r^2} + \lambda_\theta \frac{v_t}{r^2} \\ \dot{\lambda}_{v_r} & = & -\lambda_r + \lambda_{v_t} \frac{v_t}{r} \\ \dot{\lambda}_{v_t} & = & -2\lambda_{v_r} \frac{v_t}{r} + \lambda_{v_t} \frac{v_r}{r} - \frac{\lambda_\theta}{r} \end{array} \right. \quad (43)$$

In particular, from equation (27) four additional differential equations are obtained, one for each adjoint variable. In contrast, the set of differential equations depicted in (43) lacks the equations for λ_ξ . In fact, the adjoint equation for λ_ξ is:

$$\dot{\lambda}_\xi = -\frac{\partial H}{\partial \xi} = +\frac{u_T}{\xi^2} (\lambda_{v_r} \cos \phi + \lambda_{v_t} \sin \phi) = -\frac{u_T}{\xi^2} k_2 < 0 \quad (44)$$

Moreover, from equation (31), $\lambda_{\xi,f} = 0$. The equation for λ_ξ is not necessary for solving the set of differential equations and it can be computed a posteriori. Removing this trivial adjoint equation simplifies the boundary-value problem without affecting the extremals. Also, it can be noted that $\lambda_\xi(t) \geq 0 \quad \forall t$, coherently with a decreasing or stationary value of the mass ratio ξ over time.

The initial (boundary) condition are:

$$0 = r(0) - \frac{p_0}{1 + e_0 \cos \theta_0^*} \quad (45)$$

$$0 = \theta^*(0) - \theta_0^* \quad (46)$$

$$0 = v_r(0) - \sqrt{\frac{\mu}{p_0}} e_0 \sin \theta_0^* \quad (47)$$

$$0 = v_t(0) - \sqrt{\frac{\mu}{p_0}} (1 + e_0 \cos \theta_0^*) \quad (48)$$

$$0 = \xi(0) - 1 \quad (49)$$

$$0 = \lambda_r(0) - \frac{(1 + e_0 \cos \theta_0^*)^2}{p_0 \sin \theta_0^*} \sqrt{\frac{\mu}{p_0}} (\lambda_{v_t,0} \sin \theta_0^* - \lambda_{v_r,0} \cos \theta_0^*) \quad (50)$$

$$0 = \lambda_{vr}(0) - \lambda_{vr,0} \quad (51)$$

$$0 = \lambda_{vt}(0) - \lambda_{vt,0} \quad (52)$$

where: θ_0^* , $\lambda_{vr,0}$, $\lambda_{vt,0}$ are unknown initial values; condition (45) - (49) are taken from the conditions (8) - (15); condition (50) is obtained by equation (29); conditions (51), (52) are obtained by equation (30). In particular, the condition (50) is here derived:

$$0 = \underline{\nu}^\top \frac{\partial \underline{\psi}}{\partial \theta_0^*} = -\nu_1 \frac{p_0 e_0}{(1 + e_0 \cos \theta_0^*)^2} \sin \theta_0^* - \nu_3 \sqrt{\frac{\mu}{p_0}} e_0 \cos \theta_0^* + \nu_4 \sqrt{\frac{\mu}{p_0}} e_0 \sin \theta_0^* \quad (53)$$

From equation (50):

$$\lambda_{r,0} = -\nu_1 \quad \lambda_{v_r,0} = -\nu_3 \quad \lambda_{v_t,0} = -\nu_4 \quad (54)$$

Finally, it follows:

$$\lambda_{r,0} \frac{p_0 e_0}{(1 + e_0 \cos \theta_0^*)^2} \sin \theta_0^* + \lambda_{v_r,0} \sqrt{\frac{\mu}{p_0}} e_0 \cos \theta_0^* - \lambda_{v_t,0} \sqrt{\frac{\mu}{p_0}} e_0 \sin \theta_0^* = 0 \quad (55)$$

This equations yields the dependency of one of the adjoint variable on the others. Therefore, one of them, λ_r in this case, can be obtained by the guessed values of the other adjoint variables.

The final (boundary) condition are:

$$0 = r(t_f) - r_f \quad (56)$$

$$0 = v_r(t_f) - 0 \quad (57)$$

$$0 = v_t(t_f) - \sqrt{\frac{\mu}{r_f}} \quad (58)$$

$$0 = \lambda_r^2 + \lambda_{v_r}^2 + \lambda_{v_t}^2 - 1 \quad (59)$$

where: t_f is unknown; condition (56) - (58) are among the conditions (8) - (15); condition (59) replaces condition (33). In particular, in the application of Pontryagin's Maximum

Principle, the Euler-Lagrange equations and necessary conditions for optimality are homogeneous in $\underline{\lambda}$: if $(\underline{x}(t), \underline{\lambda}(t), \underline{u}(t))$ satisfies them, then for any nonzero scalar k_3 the triple $(\underline{x}(t), k_3 \underline{\lambda}(t), \underline{u}(t))$ also satisfies them, because both the Hamiltonian dynamics and the minimization depend only on the direction of $\underline{\lambda}$, not its magnitude. Consequently, $\underline{\lambda}$ is determined only up to a nonzero constant factor, and one must introduce an auxiliary normalization to select a unique representative of this equivalence class. A convenient choice is represented by:

$$\|\underline{\lambda}(t_f)\| = 1 \quad (60)$$

This normalization does not alter the validity of the PMP conditions, it simply removes the scale indeterminacy and renders the resulting two-point boundary-value problem well-posed. In addition, imposing that all the adjoint can be unity at maximum enhances the numerical conditioning of the algorithm.

Finally, the optimization process aims to find the optimal value for the variables $Y_P := \{\theta_0^*, \lambda_{vr,0}, \lambda_{vt,0}, t_f\}$, that complete the set of initial (boundary) conditions. The values are found when, evolving the initial state through the dynamical equations, the resulting final (boundary) conditions are met within a certain tolerance.

4 Implementation

Two different solvers have been used to compute the optimal solution.

The first uses MATLAB's built-in `fmincon`, where the objective

$$J = t_f \quad (61)$$

is minimized subject to linear constraints – in particular the terminal boundary conditions and the normalization $\|\underline{\lambda}(t_f)\| = 1$ – implemented in `objFun` and `nonlcon`. The solver employs sequential quadratic programming (SQP) with finite-difference gradient approximations.

The second approach is a custom shooting method: a function `shooting(YP)` takes the initial (boundary) conditions guessed and computes the residuals of the terminal conditions; a Newton–Raphson solver, including numerical Jacobian via finite differences and back-tracking (with bisection) iteratively updates the initial guess Y_P until convergence. In the implementation of a Newton-like root-finding algorithm, specific control parameters are used to guide convergence and ensure numerical stability. The parameter `RMIN`

acts as a relaxation coefficient in the Newton update formula:

$$x_{k+1} = x_k - \text{RMIN} \cdot \frac{f(x_k)}{f'(x_k)} \quad (62)$$

This modification helps to reduce the step size, preventing divergence or overshooting in regions where the function is steep or ill-conditioned. The parameter PBIS is a bisection safeguard coefficient, used when Newton's method does not converge or produces unstable steps; it allows the algorithm to fall back on a more robust bisection approach. Finally, JMAX defines the maximum number of iterations, ensuring that the algorithm terminates even if a root is not found within a reasonable number of steps.

Both algorithm steers from the following guessed values:

$$Y_P := [\theta_0^* \quad \lambda_{vr,0} \quad \lambda_{vt,0} \quad t_f]^\top = [\pi/2 \quad 0 \quad -1 \quad 60]^\top \quad (63)$$

It is worth remembering that the adjoint variables determine the change of the associated state variable. Setting a negative $\lambda_{vt,0}$, for example, enforces the search for solutions such that v_t is initially increasing (possibly beneficial for reaching outer orbits).

5 Results

The convergence parameters (e.g., residuals, iteration counts) from both implementations are reported to validate algorithmic performance. However, the graphical analysis of results — such as convergence trajectory or thrust angle — is presented solely for one implementation. This is because the outputs from the second implementation were numerically identical to the first, rendering their graphical representation redundant.

Iter	F-count	f(x)	Feasibility	First-order optimality	Norm of step
0	5	6.0000000e+01	2.642e+00	1.474e-14	
1	14	5.903887e+01	7.191e-01	1.929e-14	1.600e+00
2	19	4.913595e+01	6.234e-01	4.010e+01	9.911e+00
3	27	4.945047e+01	5.459e-01	7.639e+01	4.586e-01
4	34	5.049640e+01	3.734e-01	2.337e+02	1.135e+00
5	39	5.528254e+01	2.251e-01	3.684e+02	4.826e+00
6	44	5.569303e+01	5.828e-02	3.957e+00	4.379e-01
7	49	5.568942e+01	3.060e-03	3.355e+00	6.267e-03
8	54	5.568937e+01	8.912e-06	1.113e+00	1.061e-04
9	59	5.568937e+01	5.767e-11	8.386e-04	7.359e-07
10	64	5.568937e+01	3.094e-12	5.807e-05	4.562e-12

Local minimum possible. Constraints satisfied.

fmincon stopped because the size of the current step is less than the value of the step size tolerance and constraints are satisfied to within the value of the constraint tolerance.

<stopping criteria details>

```

--- Optimized Parameters ---
Initial true anomaly θ*(0)           : 145.761 deg
Costate λ_vr(0) (radial velocity)    : -1.890e-02 TU4/DU
Costate λ_vt(0) (tang. velocity)     : -1.042e+00 TU4/DU
Optimized time of flight tf          : 55.689 Time Unit

```

Figure 3: Convergence history and optimization results from `fmincon` solver. The table shows 14 iterations with decreasing residual norms ($\|f(x)\|$) from 6.0×10^1 to 5.1174×10^1 , achieving a final feasibility of 1.195×10^{-11} . The solver terminated due to step size (1.967×10^{-10}) and constraint tolerance satisfaction. The dimension of velocities costate are referred to as s^2/m but indicating TU^2/DU .

```

Iter: 126
YP: 2.5440e+00 -1.8904e-02 -1.0417e+00 5.5689e+01
ER: 1.2008e-12 -4.9496e-15 -2.9071e-13 -1.7941e-13
Iter: 127
YP: 2.5440e+00 -1.8904e-02 -1.0417e+00 5.5689e+01
ER: 1.2390e-12 8.6819e-14 -1.6875e-13 4.9294e-14
Iter: 128
YP: 2.5440e+00 -1.8904e-02 -1.0417e+00 5.5689e+01
ER: 1.2932e-12 1.0181e-13 -1.3878e-13 -1.0059e-13
Iter: 129
YP: 2.5440e+00 -1.8904e-02 -1.0417e+00 5.5689e+01
ER: -9.9476e-13 -1.8039e-13 -1.5593e-13 1.2323e-13
Iter: 130
YP: 2.5440e+00 -1.8904e-02 -1.0417e+00 5.5689e+01
ER: 1.8856e-12 2.0998e-13 -5.1015e-14 4.9294e-14

--- Optimized Parameters ---
Initial true anomaly  $\theta^*(0)$  : 145.761 deg
Costate  $\lambda_{vr}(0)$  (radial velocity) : -1.890e-02 TU2/DU
Costate  $\lambda_{vt}(0)$  (tang. velocity) : -1.042e+00 TU2/DU
Optimized time of flight tf : 55.689 Time Units

```

Figure 4: Final convergence state of the custom shooting method after 137 iterations. The solution achieves residuals $\mathcal{O}(10^{-12} - 10^{-14})$ with identical optimized parameters to `fmincon`. The dimension of velocities costate are referred to as s^2/m but indicating TU^2/DU .

The numerical optimization results demonstrate equivalent solutions from both approaches despite different convergence paths. While `fmincon` (Fig. 3) converged in 14 iterations, the shooting method (Fig. 4) required 137 iterations to reach residuals ($\|E_R\| < 10^{-12}$).

The two methods converge to the same solution. The optimized values for the initial guess are:

$$Y_P := [\theta_0^* \quad \lambda_{vr,0} \quad \lambda_{vt,0} \quad t_f]^\top = [145.761 \quad -1.890 \quad -1.041 \quad 55.689]^\top \quad (64)$$

The optimal time of flight results to be 55.689 TU .

The optimal transfer trajectory is shown in Figure 5. The algorithm succeeds in finding a transfer orbit that leaves the departure orbit and approaches the target orbit with only transverse velocity (to insert in a circular orbit). A overall insight shows that the constraint about the final radius and final radial velocity are met. A closer insight is given in the next figures.

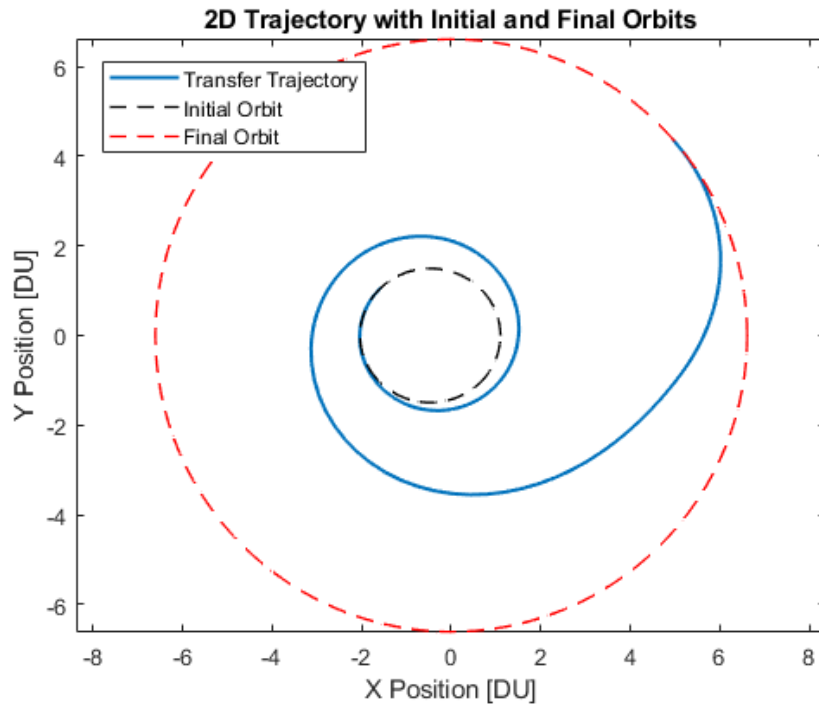


Figure 5: Optimized transfer trajectory (blue) between initial (black) and final (red) orbits.

Figure 6 shows the state variables r , θ^* , v_r and v_t . It can be noticed that the final radius boundary condition is met ($r_f \approx 6.61 \text{ DU}$) and the final radial velocity is zero, which also meets the final boundary conditions.

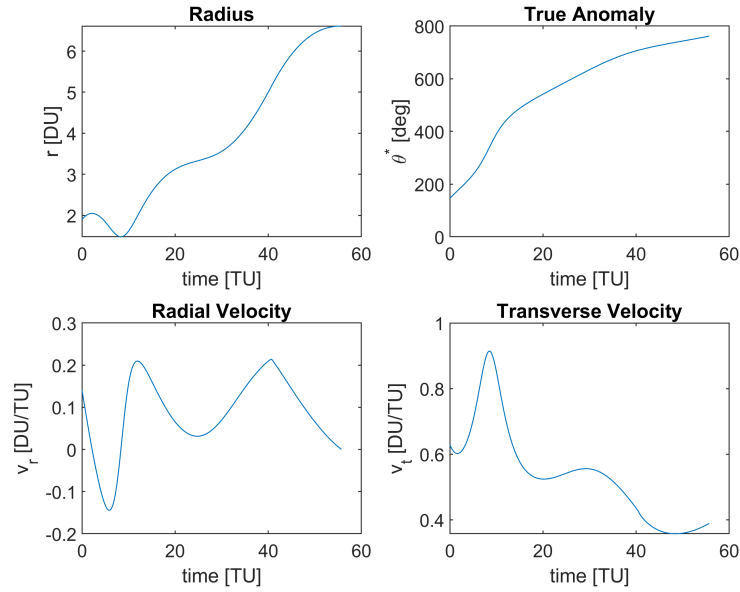


Figure 6: Orbital state variables evolution: (up left) radius, (up right) true anomaly, (bottom left) radial velocity and (bottom right) transverse velocity.

A better visualization is given in Figure 7 that shows the difference between the instantaneous value of some of the state variables with the related target value. This visualization clearly proved that the final boundary conditions (56) - (58) are met since the error flattens for all the state variables involved in the boundary conditions.

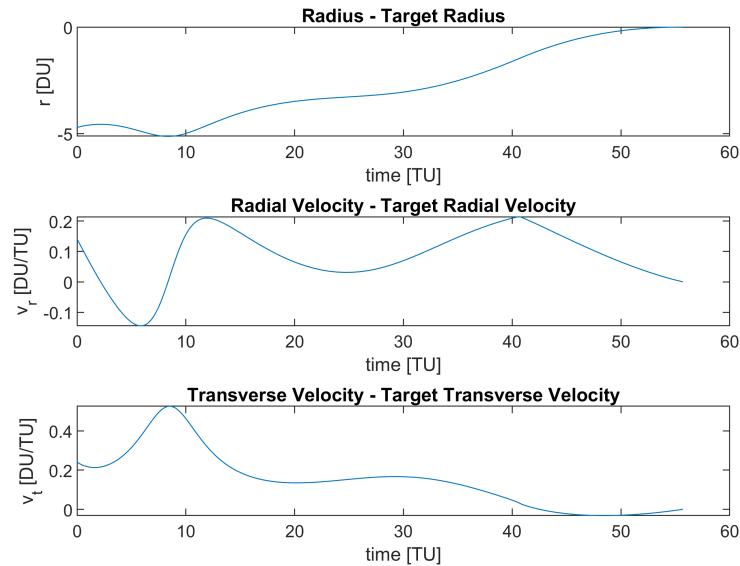


Figure 7: State variable errors: (up) Radius error, (middle) Radial velocity error, and (bottom) Transverse velocity error.

Figure 8 shows the evolution of the Tait-Bryant angles of the body-fixed commanded axes during the maneuver. As expected, since the planar nature of the transfer orbit, the

only varying angle is the yaw angle, i.e. the angle about the local vertical axis of the spacecraft. The pitch and roll angles are constant.

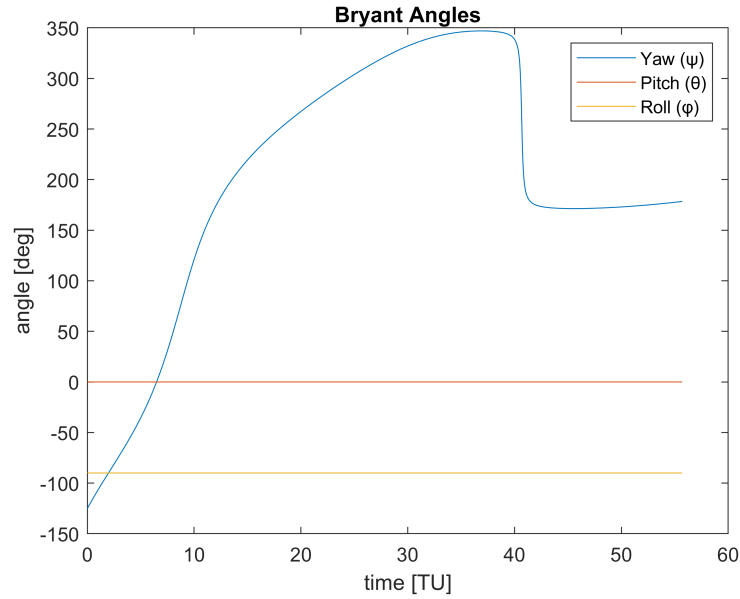


Figure 8: Bryant angles evolution during the maneuver: yaw ψ , pitch θ , and roll ϕ components over mission time.

Figure 9 shows the evolution of ϕ , the angles between the thrust direction and the radial direction. At the beginning of the maneuver, the thrust angle is kept at about 90, that is, a thrust in the velocity direction that yields the increment of the semi-major axis and eccentricity of the instantaneous osculating orbital parameters. Also in the final moment of the maneuver, the angle approaches $-270 = 90$, in order to circularize the orbit.

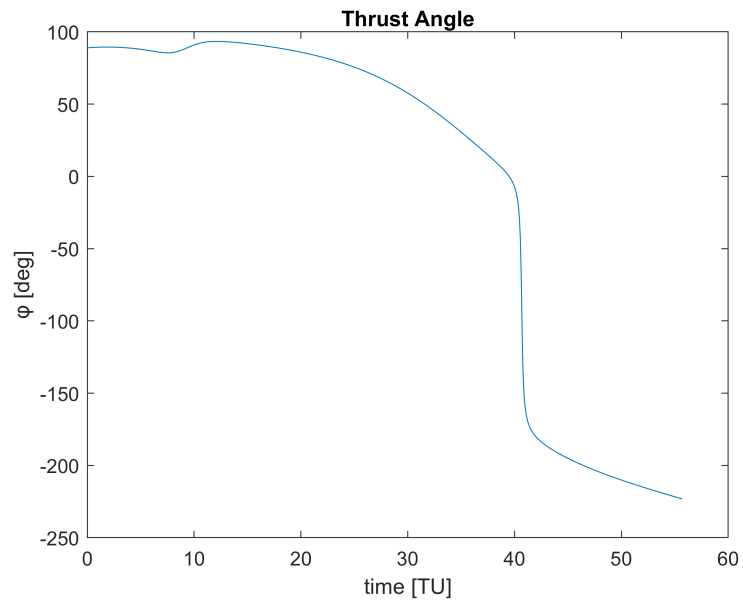


Figure 9: Optimal thrust vector orientation profile.